

681 Notes 1

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1 Introduction

Let $X \times Y$ have a topology $\mathcal{T}_{X \times Y}$ such that for all spaces Z and $f : Z \rightarrow X, g : Z \rightarrow Y$ cont, then $f \times g : Z \rightarrow X \times Y$ is cont, and $\pi_1 : X \times Y \rightarrow X, \pi_2 : X \times Y \rightarrow Y$ are cont. Then $\mathcal{T}_{X \times Y} = \mathcal{T}_P$, where $\mathcal{T}_P$ is the usual product topology on $X \times Y$.

If π_1, π_2 are continuous, for all $U \in \mathcal{T}_X, \pi_1^{-1}(U) \in \mathcal{T}_{X \times Y}$ and similarly for $V \in \mathcal{T}_Y, \pi_2^{-1}(V) \in \mathcal{T}_{X \times Y}$. $\mathcal{T}_{X \times Y}$ contains a subbasis for $\mathcal{T}_P$, so $\mathcal{T}_P \subset \mathcal{T}_{X \times Y}$.

Take $Z = X \times Y$ with the topology $\mathcal{T}_P$. Then π_1, π_2 are cont, and $f \times g = \text{id}_{X \times Y}$ is cont by assumption, so $\mathcal{T}_{X \times Y} \subset \mathcal{T}_P$, so $\mathcal{T}_{X \times Y} = \mathcal{T}_P$.

Now if $\mathcal{T}_{X \times Y} = \mathcal{T}_P$, it can be shown that all of the properties are satisfied (cont of π_1, π_2 were already implicitly assumed).

If $A \subset X$ and $i : A \hookrightarrow X$ is continuous, then for all $U \in \mathcal{T}, i^{-1}(U) = U \cap A \in \mathcal{T}_A$, so $\mathcal{T}_S \subset \mathcal{T}_A$. $\mathcal{T}_A$ is determined if you take it to be the coarsest.

Let $\bar{S}$ be the topologist sine curve, V the vertical component. Claim: $\bar{S}$ is not path connected.

Pick $x_0 = (0, 0)$ and $x_1 \in S$ and let $\gamma : I \rightarrow \bar{S}$ be continuous st $\gamma(0) = x_0, \gamma(1) = x_1$. Let $A = \gamma^{-1}(V)$, closed and contained in I . Let $l \in A$ be maximal. Note $l \neq 1$ otherwise $\gamma(l) \in V$ and $\gamma(1) \in S$. Take the homeomorphism $[l, 1] \rightarrow [0, 1]$, so let $\gamma : I \rightarrow \bar{S}$ such that $x(0) = 0$ and $x(t) > 0$ for $t > 0$.

Pick $0 < r_n < x(1/n)$ such that $\sin \frac{1}{r_n} = (-1)^n$. Pick $0 < t_n < 1/n$ such that $x(t_n) = r_n$. Then $t_n \rightarrow 0$ but $y(t_n) = \sin \frac{1}{x(t_n)} = (-1)^n$, so y is not continuous.